

PROFILE

Software Engineer at Nielsen building backend services, data pipelines, and CI/CD infrastructure. Backend-focused with full-stack capability, strong DevOps practices (Docker, Kubernetes, GitLab CI/CD), and keen interest in microservices and cloud-native deployment. Experienced in Agile/Scrum, Git branching, code reviews, and prod releases.

EXPERIENCE

Software Engineer — Nielsen

Mar 2025 – Present | Gurugram

Unified Schema — Platform powering AskNielsen analytics (USA + 28 countries)

- Built Python SDK client libraries for internal Data Lake and Trino APIs — adopted as team standard for schema registration, data ingestion, and query execution
- Designed data validation framework (pytest) with 5 validator types running via scheduled Airflow DAGs across 29 countries
- Built observability logging (SQLAlchemy + PostgreSQL/RDS) tracking SQS events, DAG states, and validation results
- Developed SQS event listener DAG triggering country-specific ETL pipelines; implemented failure alerting via email and Google Chat webhooks

NTS — Kubernetes CronJob → Airflow Migration & Big Data Pipeline Migration

- Migrated daily sync-check scripts from K8s CronJobs to Airflow (KubernetesPodOperator); added additional sync checks and Google Chat alert enhancements
- Set up GitLab CI/CD for Docker builds and Container Registry; configured cross-account S3 via IAM assume role and ensured security of service account credentials and environment variables
- Migrated 6 weekly TV rating data files to new format by modifying legacy shell scripts on production Linux server
- Validated 200K+ records ensuring correctness of migration for smooth transition to new data formats, eliminating a manual weekly workflow

PROJECTS

Microservices Chat Platform | Node.js, TypeScript, Express, Redis, RabbitMQ, Docker, K8s, GitLab CI/CD

- 4-service event-driven system (Auth, User, Chat, Gateway) — RabbitMQ async messaging, database-per-service (MySQL, PostgreSQL, MongoDB), Redis caching for conversations
- Shared common package (Zod validation, JWT middleware, inter-service auth via X-Internal-Token headers); deployed on k3d K8s (3 nodes, NGINX Ingress, multi-env namespaces)
- Full GitLab CI/CD — 4 stages, parallel builds, commit SHA tagging, automated health checks

Patient Management System | Node.js, TypeScript, gRPC, Kafka, Protobuf, Prisma, Docker, K8s + Python/FastAPI

- 5-service system (Auth, Patient, Billing, Analytics, Gateway) — gRPC for low-latency synchronous billing, Kafka for async event streaming to Analytics
- Each service with own PostgreSQL (Prisma), multi-stage Docker builds, Swagger docs, Vitest integration tests; deployed on k3d K8s with NGINX Ingress as single entry point
- Full project duplicated in Python (FastAPI) — same architecture, same deployment, proving language-agnostic infra design

Hushh Personal Agent | Next.js, TypeScript, PostgreSQL, Prisma, LangChain, Ollama/OpenAI

- Full-stack AI assistant with two AI personas, streaming chat (Ollama/OpenAI), encrypted vault, and consent-based access
- Backend API routes with Prisma-PostgreSQL schema, user authentication, LLM action parsing, and prompt engineering — developed via AI-assisted development workflows (Kiro IDE)

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL, C++

Architecture: Microservices, API Gateway, RabbitMQ, Kafka

Cloud & DevOps: AWS, Docker, Kubernetes, GitLab CI/CD

Frontend: React, Redux, React Query, Next.js (AI-assisted)

Backend: FastAPI, Express.js, Next.js, REST, gRPC, GraphQL

Databases: PostgreSQL, MongoDB, MySQL, Redis

Orchestration: Apache Airflow, Event-driven ETL

Tools: Git, Linux, pytest, Prisma, SQLAlchemy

EDUCATION

Lovely Professional University — B.Tech CSE | CGPA: 9.03
Core CS | Minor: Full Stack Web Dev | Open Minor: DevOps
The Air Force School — XII: 86% | KV AFS Rajokri — X: 96.6%

CERTIFICATIONS

- NPTEL Cloud Computing
- AWS Cloud Practitioner CLF-C02 — Udemy
- Docker & Kubernetes — Udemy